3.3 Assumptions and Constraints

- The decoded information depends on the completeness and quality of received PingStation frames.
- The project uses local storage and is intended for academic/project-level monitoring, not certified operational surveillance.
- Map visualization depends on browser access to map tile resources.
- Prediction rows are based on recent stored movement information and are limited by configured age and sample limits.

3.4 Data Dictionary

The following data dictionary lists the most important fields used throughout the backend and dashboard. Keeping these names consistent helps the frontend, storage layer and export files use the same interpretation of aircraft data.

Table 9: Data Dictionary

Field	Meaning	Unit / Format
icaoAddress	Unique aircraft ICAO address when available.	Hexadecimal text
callsign	Flight callsign or aircraft label.	Text
latDD	Latitude in decimal degrees.	Degrees
lonDD	Longitude in decimal degrees.	Degrees
altitudeMM	Aircraft altitude normalized by the preprocessor.	Millimetres
headingDE2	Heading normalized as degrees multiplied by 100.	Degrees x 100
horVelocityCMS	Horizontal speed normalized by the preprocessor.	Centimetres/second
timestamp	Observation time used for storage and replay.	ISO date-time
flight_key	Internal key used to group records for one flight.	Text

3.5 API Specification

Table 10: API Specification

Endpoint	Method	Purpose
/	GET	Render the main dashboard page.
/api/aircraft	GET	Return normalized live aircraft records as JSON.